

Dheer Guda

gudadheer@gmail.com | (732)-268-0687 | linkedin.com/in/dheer-guda | dheerguda.com

EDUCATION

University of Maryland College Park

B.S. in Computer Science & Math, Minor in Business Analytics & Data Science

College Park, Maryland

Expected Graduation, May 2027

- **Honors:** Dean's List, QUEST Honors Program, Startup Shell Grant Winner
- **Related Coursework:** Data Structures & Algorithms, Operating Systems, Database Design, Machine Learning, Artificial Intelligence, Software Engineering, Probability & Statistics

EXPERIENCE

Software Development Engineer Intern, Amazon

Seattle, Washington

LLM Deployment Cost Optimization (MARS)

- Extended MARS, Amazon's internal system for finding a model's cheapest deployment configuration, beyond weight-only optimizations (e.g. quantization) by adding knowledge distillation, its first parameter-modifying technique.
- Built hard-label, soft-label, and feature-based distillation pipelines with NVIDIA Model-Opt, Axolotl, and LLaMA-Factory to fine-tune the production Qwen 3.5 model that fills missing product type attributes (PTA) on Amazon's catalog.
- Designed a pipeline-composition framework spanning Model-Opt techniques (speculative decoding, pruning, quantization, distillation), using heuristic and surrogate-model search to find the best-serving combination.

CEO, Synari

College Park, Maryland

AI-Powered Practice Management Startup for Cognitive Therapy (synari.org)

- Grew platform to \$1,000 monthly recurring revenue (MRR) within 3 months by acquiring paying therapist clients.
- Built and launched core features (billing, treatment planning, clinical notes) with Python, FastAPI, React, and Stripe, reducing documentation time for therapists by 40%. Built and deployed with Docker on Railway through Git.
- Implemented secure HIPAA-compliant AWS infra (EC2, S3, RDS) with encryption, IAM, and VPC isolation.
- Selected for Telora Fellowship and engaged with Mokhtarzada Hatchery, refining strategy and pitching to investors.

Software Engineer, University Career Center (UCC)

Remote, New Jersey

Full-Stack Software Engineering (Academic-Year Internship)

- Built and maintained Intern for a Day (ifad.umd.edu) serving 1,000+ students/cycle on FastAPI, React/TS, PostgreSQL.
- Built custom student matching algorithm integrating Google Gemini for semantic checks, halving coordinator time.
- Led a \$35K Do Good-funded upgrade adding RBAC, FERPA-aware PII controls, audit logs, and data-retention policies.
- Shipped CI/CD (GitHub Actions + Docker) and monitoring; query tuning and caching cut p50 page load ~30%.

Startup Software Engineer Intern, SPARK

Remote, New Jersey

Full Stack Software Development Internship

- Built and deployed scalable web applications using React, Python, and FastAPI, improving UX load time by 35%.
- Integrated Stripe, AWS Lambda, and ElevenLabs voice API for secure payments, cutting latency by 20%.
- Designed backend microservices and REST APIs using PostgreSQL and Docker to support concurrent user sessions.

SAR Synthetic Dataset Research Intern, Professor Triet Le

College Park, Maryland

Software/ML Research Assistant

- Developed SAR dataset in MATLAB/Python/C to reduce data scarcity and enable self-supervised learning.
- Implemented a preprocessing pipeline (calibration, speckle filtering, normalization, tiling, augmentation) to improve model signal-to-noise and training stability.
- Benchmarked CNN baselines against classical ML (SVM/Random Forest) with k-fold CV on SAR classification, segmentation, and backscatter-regression tasks.

PROJECTS

Audit.AI (Bitcamp 2024 Hackathon Winner)

College Park, Maryland

Full Stack Developer

- Built a YOLOv6-based multimodal ML system using Python and PyTorch to detect misinformation at scale.
- Engineered scalable backend with MongoDB and Redis caching for real-time media auditing, earning best overall hack.

askTestudo

Georgetown, Maryland

Lead Backend Developer

- Trained an LLM chatbot with Azure OpenAI & Retrieval-Augmented Generation (RAG) for precise query handling.
- Integrated vector search indexing, multimodal video responses, and chat UI in a React/FastAPI/MongoDB app.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, R, HTML, CSS

Frameworks & Libraries: React.js, Node.js, FastAPI, Flask, React Native, PyTorch, TensorFlow, Bootstrap

Cloud & DevOps: AWS, Azure, Google Cloud, Docker, Kubernetes, Cloudflare, GitHub Actions, CI/CD, Linux, Ubuntu

Databases & Tools: PostgreSQL, MongoDB, Redis, MySQL, Jupyter, Colab, VS Code, IntelliJ, Git, Eclipse, Bash